

Modelling of Stochastic Processes

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1 Introduction

What is a model (modelling)?

Phenomenon $\rightarrow$ Mathematical Problem $\rightarrow$ Mathematical solution $\rightarrow$ Interpretation $\rightarrow$ Phenomenon . . .

There are several objectives of simulations:

- Fitting the model, determining optimal parameters
- Determining the state of the system
- Numerical experiment, analysis and predictions
- Inverse problem: determining the system initial conditions

Models are classified into

- Discrete
- Continuous

They are further classified into:

- Deterministic
- Stochastic
- Heuristic (Genetic algorithms, Neural networks, Simulated annealing)

And they can be Detailed or Phenomenological.

2 Probabilities

2.1 Basics

The following are some basic properties of probabilities:

Probability of an event $P(A)$

Probability of a union of two events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Independent events

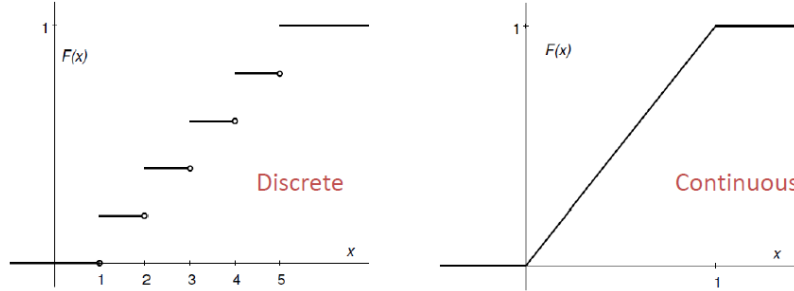
$$P(A \cap B) = P(A) \cdot P(B)$$

Bayes' rule

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

The value of the random variable X depends on the specific outcome result of stochastic events. If the range of X values is discrete then it's a **discrete random variable** and if it's continuous then it's a **continuous random variable**. Probability mass function (weighted probabilities): $p_n = \Pr\{X = x_n\}$. Note that $\sum_n p_n = 1$.

Cummulative distribution function $F(x) = \Pr\{X \in (-\infty; x]\}$.



Probability density function $P_X = \frac{dF(x)}{dx}$, for a discrete random variable $P_X(x) = \sum_n p_n \delta(x - x_n)$. So the probability is just an integral

$$\Pr\{a \leq X \leq b\} = \int_a^b P_X(x) dx = F(b) - F(a)$$

The n -th moment of X is an expectation (mean) value of X^n

$$E(X^n) \equiv \langle x^n \rangle = \int x^n P_X(x) dx \quad \langle x^n \rangle = \sum_k x_k^n p_k$$

Expectation (mean) value

$$\mu_X = \langle x \rangle = \int x P_X(x) dx \quad \text{or} \quad \mu_X = \langle x \rangle = \sum_k x_k p_k$$

Mean value of a function $u(x)$

$$\langle u(X) \rangle = \int u(x) P_X(x) dx \quad \text{or} \quad \langle u(X) \rangle = \sum_k u(x_k) p_k$$

Note that expectation value is a linear operator.

Variance

$$\sigma_X^2 = \langle [X - \mu_X]^2 \rangle = \langle x^2 \rangle - \mu_X^2$$

Example: Bernoulli distribution

Two possible values $X = \{x_0, x_1\} = \{0, 1\}$, with probability mass function $p_0 = q = 1 - p$ and $p_1 = p$. Then

$$\mu_x = \langle x \rangle = p \quad \text{and} \quad \sigma_X^2 = \langle x^2 \rangle - \langle x \rangle^2 = p(1 - p)$$

2.2 Generating functions

Assume X is a discrete random variable $\in \mathbb{N}_0$, with probabilities

$$X = \{0, 1, 2, \dots\}; \quad \Pr\{X = n\} = p_n; \quad \sum_{n=0}^{\infty} p_n = 1$$

then

$$\mu_X = \sum_{n=0}^{\infty} n p_n; \quad \sigma_X^2 = \sum_{n=0}^{\infty} n^2 p_n - \mu_X^2$$

The **generating function** of the random variable X is defined as the following function of the real number s :

$$G_X(s) = \langle s^x \rangle = \sum_{n=0}^{\infty} s^n p_n$$

Note that the series converges for any $|s| \leq 1$.

$$G'_X(1) = \sum_n n p_n = \mu_X; \quad G''_X(1) = \sum_n n(n-1) p_n = \langle n^1 \rangle - \mu_X$$

Also we can retrieve probabilities from the generating function

$$G_X(1) = 1 \quad G_X(0) = p_0 \quad G'_X(0) = p_1 \quad G''_X(0) = 2! p_2 \quad \dots$$

For the continuous case we have

$$G_X(s) = \int P_X(x) s^x dx, \quad |s| \leq 1$$

Similarly, Mean value and Variance are given by

$$\mu_X = G'_X(1) \quad \sigma_X^2 = G''_X(1) + \mu_X - [\mu_X]^2$$

For Independent random variables X_1 and X_2

$$G_{X_1+X_2}(s) = \langle s^{x_1+x_2} \rangle = \langle s^{x_1} \rangle \langle s^{x_2} \rangle = G_{X_1}(s) \cdot G_{X_2}(s)$$

Example: Bernoulli distribution

$$X = \{x_0, x_1\} = \{0, 1\} \quad \rightarrow \quad p_n = \{1 - p, p\}$$

- Mean Value: $\mu_X = \langle x \rangle = p$
- Variance $\sigma_X^2 = \langle x^2 \rangle - \mu_X^2 = p(1 - p)$
- Generating function

$$G_X(s) = \sum_{n=0}^1 s^n p_n = (1 - p) + sp$$

2.3 Example: Uniform distribution

Uniform discrete distribution: $X = \{1, 2, \dots, N\}$, $p_k = \frac{1}{N}$

$$\mu_X = \frac{N+1}{2} \quad \sigma_X^2 = \frac{N^2-1}{12}$$

Generating function

$$G_X(s) = \sum_{k=1}^N s^k p_k = \frac{1}{N} \sum_{k=1}^N s^k = \frac{s}{N} \frac{1-s^N}{1-s}$$

Uniform continuous distribution: $X \in [0; 1]$, $P_X(x) = 1$

$$\mu_X = \frac{1}{2}, \quad \sigma_X^2 = \frac{1}{12}$$

$$G_X(s) = \int_0^1 s^x dx = \frac{s-1}{\ln s}$$

Example: Binomial distribution

Assume a population of N bacteria, each of them can be either alive (with probability p) or dead (with probability $1 - p$)

Let $Y = \{0, 1, 2, \dots\}$ be the number of alive bacteria, then

$$p_k = \frac{N!}{k!(N-k)!} p^k (1-p)^{N-k}$$

Note that $Y = X_1 + X_2 + \dots + X_N$, where X_j are independent Bernoulli's variables, then this makes it easier to compute the generating function

$$G_Y(s) = [G_{X_1}(s)]^N = [1 - p + sp]^N$$

Then mean value and variance are just

$$\mu_Y = G'_Y(1) = Np \quad \text{and} \quad \sigma_Y^2 = G''_Y + \mu_Y - \mu_Y^2 = Np(1-p)$$

Example: Poisson distribution

It corresponds to the binomial distribution in the limit $N \rightarrow \infty$, $p \rightarrow 0$, with the product $Np = \lambda = \text{const.}$

$$p_k = \frac{\lambda^k e^{-\lambda}}{k!}$$

Generating function

$$G_X(s) = \sum_{k=0}^{\infty} s^k p_k = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(s\lambda)^k}{k!} = \exp(\lambda(s-1))$$

Again the mean value and variance can be easily obtained

$$\mu_X = G'_X(1) = \lambda \quad \text{and} \quad \sigma_X^2 = G''_X(1) + \mu_X - \mu_X^2 = \lambda$$

Exponential distribution

The distribution of the delay times until some event is usually described by the exponential law:

$$X \in [0; \infty), \quad P_X(x) = \lambda e^{-\lambda x},$$

$$\mu_X = \int_0^{\infty} \lambda x e^{-\lambda x} dx = \frac{1}{\lambda} \quad \sigma_X^2 = \int_0^{\infty} \lambda x^2 e^{-\lambda x} dx - \mu_X^2 = \frac{1}{\lambda^2}$$

Example: Normal (Gaussian) distribution

$$X \in (-\infty; \infty), \quad P_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\mu_X = \mu \quad \sigma_X^2 = \sigma^2$$

2.4 Stochastic processes

A stochastic process is a sequence of random variables when the value of the random variable X changes with time. Assume that at some specific time moments $t_1, t_2, t_3 \dots$ the random variable X acquired values X_1, X_2 and so on respectively. Then the multi-variable probability density function $P(X_1, t_1; X_2, t_2; \dots)$. And the conditional probability

$$P(X_1, t_1; X_2, t_2; \dots | X'_1, t'_1; X'_2, t'_2; \dots) = \frac{P(X'_1, t'_1; X'_2, t'_2; \dots; X_1, t_1; X_2, t_2; \dots)}{P(X'_1, t'_1; X'_2, t'_2; \dots)}$$

The Markov chain is a stochastic process *without memory* meaning

$$P(X_1; t_1; X_2, t_2; \dots | X'_1, t'_1)$$

Thus for some time series

$$P(X_1, t_1; X_2, t_2; \dots X_n, t_n) = \prod_{k=1}^n P(X_k, t_k | X_{k-1}, t_{k-1})$$

If the one-step probability does not depend on time, the process is called the stationary (or time-homogeneous) Markov-Chain:

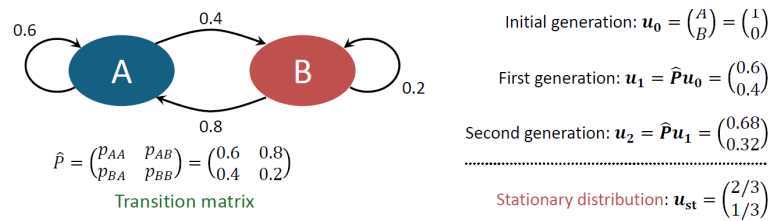
$$P(X_2, t_2 | X_1, t_1) = p_{21}$$

Where $\sum_j p_{ji} = 1$ and $p_{ji} > 0$

Example: Two species

Suppose there is a forest, where only two species of trees can grow - say, A and B. with similar growing rate and lifetime.

In general each generation, every old tree is replaced with a new young tree of either the same or different species. This is discrete-time Markov process, where time t_n represents the n -th generation



3 Markov chains

Discrete-time stochastic process $\{X_n = X(t_n)\}_{n=0}^{\infty}$ is called **Markov Process**, if $\Pr\{X_n = i_n | X_{n-1} = i_{n-1}\}$, here i_k belongs to the state space of the random variable X . Of this state space of X is **finite** then it's a **finite Markov Chain**. **One-step transition probability** $p_{ji}(n) = \Pr\{X_{n+1} = j | X_n = i\}$. If $p_{ji}(n) \equiv p_{ji} = \text{const}$, the Markov chain is **stationary**. Note that $\sum_j p_{ji} = 1$.

We can define a transition matrix

$$P = \begin{pmatrix} p_{11} & p_{12} & p_{13} & \cdots \\ p_{21} & p_{22} & p_{23} & \cdots \\ p_{31} & p_{32} & p_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

The sum is equal to 1
(stochastic matrix)

If the sum is also equal to 1
→ doubly stochastic matrix

If P is stochastic matrix, then P^n is a stochastic matrix as well. If P is doubly stochastic matrix, then P^n is a doubly stochastic matrix as well.

We can also define n -step transition probability i.e. the probability of transferring from state i to state j in n time steps

$$p_{ji}^{(n)} = \Pr\{X_n = j | X_0 = i\}$$

Then clearly

$$\begin{aligned} n = 0 & \Rightarrow p_{ji}^{(0)} = \delta_{ji} \\ n = 1 & \Rightarrow p_{ji}^{(1)} = p_{ji} \end{aligned}$$

3.1 Chapman-Kolmogorov equation

The Chapman-Kolmogorov equation relates the n -step transition probability with the s -step and $(n-s)$ -step transition probabilities:

$$p_{ji}^{(n)} = \Pr\{X_n = j | X_0 = i\} \quad (1)$$

$$= \sum_{k=1}^{\infty} \Pr\{X_n = j, X_s = k | X_0 = i\} \quad (2)$$

$$= \sum_{k=1}^{\infty} \Pr\{X_n = j | X_s = k, X_0 = i\} \Pr\{X_s = k | X_0 = i\} \quad (3)$$

$$= \sum_{k=1}^{\infty} \Pr\{X_n = j | X_s = k\} \Pr\{X_s = k | X_0 = i\} \quad (4)$$

$$= \sum_{k=1}^{\infty} p_{jk}^{(n-s)} p_{ki}^{(s)}. \quad (5)$$

Thus

$$p_{ji}^{(n)} = \sum_{k=1}^{\infty} p_{jk}^{(n-s)} p_{ki}^{(s)}$$

or in a matrix form: $P^{(n)} = P^{(n-s)} \cdot P^s$, the notice that $P^{(n)} = P^n$

3.2 Classification of states

Irreducible Markov chain is such that from any state it is possible to reach any other state is some number of steps. Otherwise it is **reducible** i.e. there either exist some disjoint parts or there is a way one relation between some states.

❖ **Example:** genetic mutation

During the replication of a DNA segment, there is a non-zero probability p that any of 4 bases (A, G, C or T) is accidentally replaced by another one:

$$P = \begin{pmatrix} 1-3p & p & p & p \\ p & 1-3p & p & p \\ p & p & 1-3p & p \\ p & p & p & 1-3p \end{pmatrix} \quad \text{Doubly stochastic transition matrix}$$

Starting from any initial state (A, G, C or T), there is a possibility to reach any other state (in one or several steps)

➡ Irreducible Markov chain and irreducible transition matrix

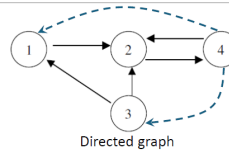
Communicating states are those who can reach one another in both directions, hence the notation $i \xrightarrow{j}$. Likewise, **communication classes** are just sets of all the states that can communicate. If there exists multiple more than one communication class then the state is reducible.

Classification of states

❖ **Example:**

$$P = \begin{pmatrix} 0 & 0 & p_{13} & 0 \\ p_{21} & 0 & p_{23} & p_{24} \\ 0 & 0 & 0 & 0 \\ 0 & p_{42} & 0 & 0 \end{pmatrix}$$

Transition matrix



❖ Starting from state 3 or 1, it is impossible to return to these states

❖ Three **communication classes**: $\{3\}$, $\{1\}$ and $\{2,4\}$

❖ Reducible Markov chain

❖ Starting from the set $\{2,4\}$, it is impossible to reach other states (1 or 3)

➡ The set $\{2,4\}$ is closed

What would change, if $p_{14} \neq 0$?

Two communication classes: $\{1,2,4\}$ and $\{3\}$

What would change, if $p_{34} \neq 0$?

Single communication class $\{1,2,3,4\}$

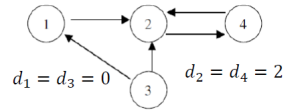
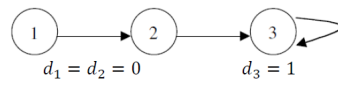
⇒ **irreducible chain**

Absorbing boundaries are such boundaries which when entered are never left i.e. $p_{ii} = 1$. If there exists absorbing boundaries, then there are several communication classes and thus the markov chain is reducible.

Period of a state d_i of state i is the greatest common divisor of all integers $n \geq 1$, for which $p_{ii}^n > 0$

- If $d_i > 1$, state i is **periodic** with period d_i
- If $d_i = 1$, state i is **aperiodic**
- If $p_{ii}^n = 0$ for all n , the period is defined as $d_i = 0$

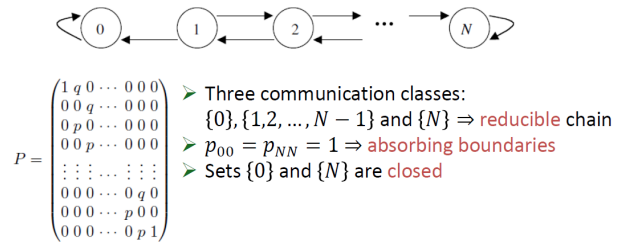
Example:



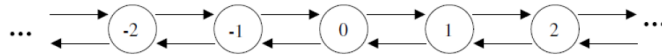
The periods of all states belonging to the same communication class are the same

❖ Example: classical gambling. The gambler bets 1€ per game and either wins 2€ with probability p or loses with probability $q = 1 - p$.

❖ The gambler starts with n € and stops playing in case of bankrupt (reaching state 0) or winning N €.

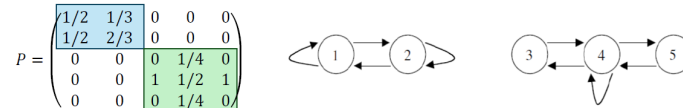


Example: Unrestricted one-dimensional random walk



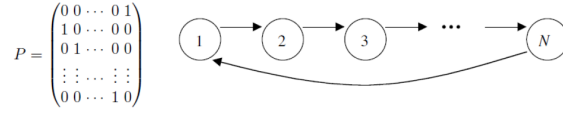
- The transition matrix is infinite
- Irreducible closed Markov chain

❖ Example:



- ❖ Two communication classes: $\{1, 2\}$, and $\{3, 4, 5\}$
- ❖ Both sets are closed
- ❖ Reducible Markov chain

❖ Example:



- ❖ Irreducible Markov chain,
the single communication class $\{1, 2, \dots, N\}$ is closed
- ❖ Starting from state i , it takes exactly N steps to return to the same state i .
Thus,

$$P^N = I.$$

- ❖ **Periodic** Markov chain, the period is equal to N

3.3 First-return time

Let f_{ii}^n denote the probability that random variable X acquired again its initial value $X_0 = i$ after n steps:

$$f_{ii}^{(n)} = \Pr\{X_n = i, X_m \neq i, m = 1, 2, \dots, n-1 | X_0 = i\}$$

Note that $f_{ii}^1 = p_{ii}$

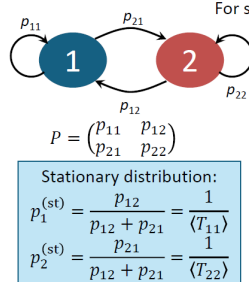
We can now define the following. **recurent** state is such a state that will be infinitely visited, but a **transient** state is such which might be visited finite number of times, but eventually it will not be visited ever again.

- If $\sum_{n=1}^{\infty} f_{ii}^n = 1$, state i is **recurent**
- If $\sum_{n=1}^{\infty} f_{ii}^n < 1$, state i is **transient**

For a recurrent state we can define a new random variable T_{ii} , whose elements $T_{ii}^n = n \in \mathbb{N}$ are acquired with probabilities $f_{ii}^{(n)}$ and is called a **first return time**. In addition, for the recurrent state, we can define the **mean recurrence** (or **first-return**) time:

$$E(T_{ii}) \equiv \langle T_{ii} \rangle = \sum_{n=1}^{\infty} n f_{ii}^{(n)}$$

Example:



For state 1: $\triangleright f_{11}^{(1)} = p_{11}, f_{11}^{(2)} = p_{12}p_{21}, f_{11}^{(3)} = p_{12}p_{22}p_{21},$
 $f_{11}^{(n)} = p_{12}p_{22}^{n-2}p_{21}, \text{ for } n \geq 3$

Is this state recurrent?

$$\triangleright \sum_{n=1}^{\infty} f_{11}^{(n)} = p_{11} + p_{12}p_{21} \sum_{n=0}^{\infty} p_{22}^n = p_{11} + \frac{p_{12}p_{21}}{1-p_{22}} = 1$$

↪ Recurrent state

$$\langle T_{11} \rangle = p_{11} + p_{12}p_{21} \sum_{n=0}^{\infty} (n+2)p_{22}^n = \frac{p_{12} + p_{21}}{p_{12}}$$

$$\triangleright \text{Analogically, } \langle T_{22} \rangle = \frac{p_{12} + p_{21}}{p_{21}}$$

Basic Limit theorem

If X_n is a recurrent, irreducible, and aperiodic Markov chain with one-step transition probabilities p_{ij} , then

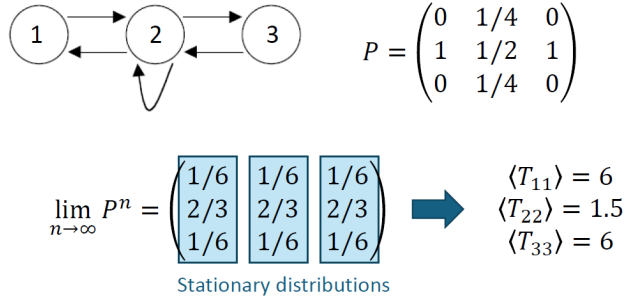
$$\frac{1}{\langle T_{ii} \rangle} = \lim_{n \rightarrow \infty} p_{ij}^{(n)} = p_i^{(\text{st})}$$

If X_n is a recurrent, irreducible, and periodic Markov chain with period $d > 1$, then

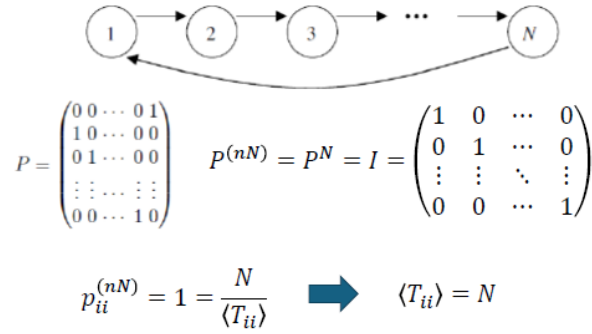
$$\lim_{n \rightarrow \infty} p_{ii}^{nd} = \frac{d}{\langle T_{ii} \rangle}$$

and $p_{ii}^{(m)} = 0$ if m is not a multiple of d .

Example: aperiodic chain



Example: periodic chain



4 Mean first-passage time

Define **first-passage probability** $f_{ji}^{(n)}$ that starting from state $X_0 = i$ after n steps you end up at state j

$$f_{ji}^{(n)} = \Pr\{X_n = j, X_m \neq j, m = 1, 2, \dots, n-1 | X_0 = i\}$$

Note that

$$0 \leq \sum_{n=1}^{\infty} f_{ji}^{(n)} \leq 1$$

Then, the mean time of first passage from state i to state j is

$$\langle T_{ji} \rangle = \sum_{n=1}^{\infty} n f_{ji}^{(n)}$$

This is more easily calculable. Define

$$T = \begin{bmatrix} \langle T_{11} \rangle & \langle T_{12} \rangle & \dots & \langle T_{1N} \rangle \\ \langle T_{21} \rangle & \langle T_{22} \rangle & \dots & \langle T_{2N} \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle T_{N1} \rangle & \langle T_{N2} \rangle & \dots & \langle T_{NN} \rangle \end{bmatrix}$$

The in matrix form we have the following identity

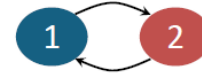
$$T = E + (T - \text{diag}(T)) \cdot P$$

where T is the matrix of mean times, E is matrix of ones (not identity, just every element is 1) and P is a transition matrix.

Example

$$T = E + (T - \text{diag}(T)) \cdot P$$

❖ Consider Markov chain with $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$



❖ The matrix equation is:

$$\begin{pmatrix} \langle T_{11} \rangle & \langle T_{12} \rangle \\ \langle T_{21} \rangle & \langle T_{22} \rangle \end{pmatrix} = \begin{pmatrix} 1 + \langle T_{12} \rangle & 1 \\ 1 & 1 + \langle T_{21} \rangle \end{pmatrix}$$

❖ Thus, we immediately get:

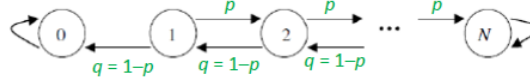
$$\begin{aligned} \langle T_{12} \rangle &= \langle T_{21} \rangle = 1 \\ \langle T_{11} \rangle &= \langle T_{22} \rangle = 2 \end{aligned}$$

5 Random Walk

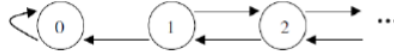
5.1 1D random walk

❖ Restricted random walk (RW):

➤ Finite: $X = \{0, 1, 2, \dots, N\}$



➤ Semi-infinite: $X = \{0, 1, 2, \dots\}$



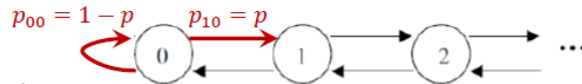
❖ Unrestricted RW: $X = \{\dots, -2, -1, 0, 1, 2, \dots\}$



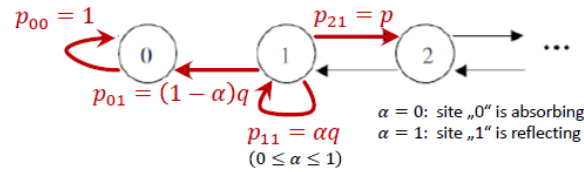
❖ Absorbing boundary:



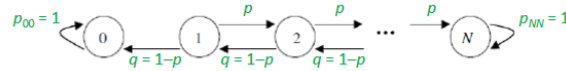
❖ Reflecting boundary:



❖ Elastic boundary:



RW with two absorbing boundaries



$$P = \begin{pmatrix} 1 & q & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & q & \dots & 0 & 0 & 0 \\ 0 & p & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & p & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & q & 0 \\ 0 & 0 & 0 & \dots & p & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & p & 1 \end{pmatrix}$$

❖ Example: Gambler, starting with initial capital k , stops the game either losing everything (reaching the state $X = 0$), or winning a jackpot (reaching the state $X = N$)

❖ What is the probability of each of these events?

❖ How long, on average, the game lasts?

$$\tau_k = \langle T_{0k} \rangle + \langle T_{Nk} \rangle$$

Define the probability to lose after n steps by starting from state k : $a_{nk} = \Pr\{X_n = 0 | X_0 = k\}$ and the probability to win after n steps: $b_{nk} = \Pr\{X_n = N | X_0 = k\}$

Then the boundary conditions are

$$a_{n0} = 1, \quad a_{nN} = 0, \quad b_{n0} = 0, \quad b_{nN} = 1$$

Combined probability to end the game after n steps is $a_{nk} + b_{nk}$, hence

$$\sum_{n=0}^{\infty} (a_{nk} + b_{nk}) = 1$$

The total probability to lose is $a_k = \sum_{n=0}^{\infty} a_{nk}$

The total probability to win is $b_k = \sum_{n=0}^{\infty} b_{nk}$

The mean duration of the game is $\tau_k = \sum_{n=0}^{\infty} n \cdot (a_{nk} + b_{nk})$

Having defined the essential variables, we can determine the relationship between a_k , a_{k-1} and a_{k+1} then

$$a_k = pa_{k+1} + qa_{k-1}$$

Which means that if you start at state k then you can only move right or left, if you move to state $k+1$ (with probability p) then your total probability of losing will be pa_{k+1} . Similarly, with moving left. We obtain a linear second-order homogeneous difference equation in terms of a_k :

$$pa_{k+1} - a_k + qa_{k-1} = 0$$

To solve this solution we always use the ansatz $a_k = \lambda^k \neq 0$

$$p\lambda^{k+1} - \lambda^k + q\lambda^{k-1} = 0 \quad \Rightarrow \quad \lambda_1 = 1 \text{ and } \lambda_2 = \frac{q}{p}$$

Losing probability

❖ 1st case: $p \neq q$

➤ General solution is $a_k = c_1 + c_2 \left(\frac{q}{p}\right)^k$

➤ From the boundary conditions: $c_1 = -\frac{(q/p)^N}{1-(q/p)^N}$, $c_2 = \frac{1}{1-(q/p)^N}$

$$a_k = \frac{(q/p)^N - (q/p)^k}{(q/p)^N - 1}, \quad p \neq q$$

$$\lambda_{1,2} = \begin{cases} 1, \\ q/p \end{cases}$$

$$a_0 = 1, a_N = 0$$

❖ 2nd case: $p = q = \frac{1}{2}$

➤ General solution is $a_k = c_1 + c_2 k$

➤ From the boundary conditions: $c_1 = 1$, $c_2 = -\frac{1}{N}$

$$a_k = \frac{N-k}{N}, \quad p = q = \frac{1}{2}$$

In both cases, the probability of winning is $b_k = 1 - a_k$

5.2 Mean duration of game

Similarly we can write an equation which ties mean duration times, but now notice that we must account for the extra step we did (since we count steps)

$$\tau_k = p(1 + \tau_{k+1}) + q(1 + \tau_{k-1}) \quad \Rightarrow \quad p\tau_{k+1} - \tau_k + q\tau_{k-1} = -1$$

This is a **nonhomogeneous** second order difference equation with boundary conditions $\tau_0 = \tau_N = 0$.

We split the solution into two parts: general homogeneous solution $\bar{\tau}_k$ and a particular solution of the nonhomogeneous equation $\tilde{\tau}_k$.

$$\tau_k = \bar{\tau}_k + \tilde{\tau}_k$$

We seek a solution of the form

$$\tau_k = \lambda^k \neq 0.$$

The characteristic equation gives

$$\lambda_{1,2} = \{1, q/p\}.$$

1st case: $p \neq q$ $\tilde{\tau}_k = c k$

$$\tau_k = c_1 + c_2 \left(\frac{q}{p}\right)^k + \frac{k}{q-p}$$

or equivalently,

$$\tau_k = \frac{1}{q-p} \left[k - N \frac{1 - \left(\frac{q}{p}\right)^k}{1 - \left(\frac{q}{p}\right)^N} \right]$$

2nd case: $p = q = \frac{1}{2}$ $\tilde{\tau}_k = c k^2$

$$\tau_k = c_1 + c_2 k - k^2 = k(N - k).$$

5.3 Probability distribution

We have determined the **total** losing and winning probabilities, but we have not determined the probabilities for separate events a_{nk} and b_{nk} . We can determine them with generating functions

$$A_k(s) = \sum_{n=0}^{\infty} a_{nk} s^n \quad \text{and} \quad B_k(s) = \sum_{n=0}^{\infty} b_{nk} s^n$$

$$a_{nk} = \frac{1}{n!} \left. \frac{d^n A_k(s)}{ds^n} \right|_{s=0}, \quad b_{nk} = \frac{1}{n!} \left. \frac{d^n B_k(s)}{ds^n} \right|_{s=0} \quad (6)$$

$$\tau_k = A'_k(1) + B'_k(1) \quad (7)$$

To derive the relations for these values we again can take the known relation for a

$$a_{n+1,k} = p a_{n,k+1} + q a_{n,k-1}$$

Multiply by s^{n+1} and sum up to obtain

$$A_k(s) = p s A_{k+1} + q s A_{k-1}$$

Again we use the same ansatz $A_k = [\lambda(s)]^k$. General solution is $A_k(s) = c_1 \lambda_1^k + c_2 \lambda_2^k$, with boundary conditions $A_0(s) = 1$ and $A_N(s) = 0$. Plugging everything in we obtain

$$A_k(s) = \frac{\lambda_1^N \lambda_2^k - \lambda_2^N \lambda_1^k}{\lambda_1^N - \lambda_2^N}$$

Semi-infinite random walk (where there is not cap on winning) can be obtained with the limit $N \rightarrow \infty$. For unrestricted random walk starting from any site, one will **always** return to this site:

$$\sum_{n=1}^{\infty} f_{00}^{(n)} = \frac{1}{2}(a_1 + a_{-1}) = 1$$

However on average the time will be infinite

$$\langle T_{00} \rangle = 1 + \tau_1 = 1 + \frac{1}{q-p} = \infty$$

6 Birth and Death Process

We can try and calculate an average RW duration

Calculation of average RW duration

$p\tau_{k+1} - \tau_k + q\tau_{k-1} = -1$

Transition matrix:

$$P = \begin{pmatrix} 1 & q & 0 & 0 & \dots & 0 \\ 0 & 0 & q & 0 & \dots & 0 \\ 0 & p & 0 & q & \dots & 0 \\ 0 & 0 & p & 0 & \dots & 0 \\ 0 & 0 & 0 & p & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 0 & p & 1 \end{pmatrix}$$

Let's denote the submatrix T

❖ From here we immediately get:

The equation can also be written in **matrix form**:

$$(\tau_1 \ \tau_2 \ \tau_3 \ \dots) \cdot \underbrace{\begin{pmatrix} -1 & q & 0 & \dots \\ p & -1 & q & \dots \\ 0 & p & -1 & \dots \\ 0 & 0 & p & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}}_{= T - I} = -(1 \ 1 \ 1 \ \dots)$$

$$(\tau_1 \ \tau_2 \ \tau_3 \ \dots) = (1 \ 1 \ 1 \ \dots) \cdot (I - T)^{-1}$$

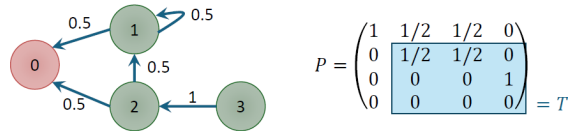
If we assume that Markov chain has k absorbing states. We can write its transition matrix in the following form:

$$P = \begin{bmatrix} I & A \\ 0 & T \end{bmatrix}$$

where I is a diagonal $k \times k$ identity matrix, 0 is a zero filled matrix. Define the fundamental matrix $F = (I - T)^{-1}$. The expected times until absorption, if random walk starts from some particular node, can be obtained by numerically multiplying two matrices.

$$(\tau_1, \tau_2, \tau_3, \dots) = (1 \ 1 \ 1 \ \dots) \cdot F$$

Example



❖ Fundamental matrix:

$$F = (I - T)^{-1} = \begin{pmatrix} 1/2 & -1/2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

❖ Row vector of the expected times until absorption:

$$(\tau_1 \ \tau_2 \ \tau_3) = (1 \ 1 \ 1) \cdot F = (2 \ 2 \ 3)$$

7 Branching Processes

Assume a random variable X_n describes the total population size after n generations. Each individual in generation n gives birth to Y_n offspring, independently from other individuals. $\Pr\{Y_n = k\} = p_k$, here $\sum_{k=0}^{\infty} p_k = 1$. If, in some generation $X = 0$, then the branching process stops.

For the random variable Y : $G_Y(s) = \sum_{k=0}^{\infty} p_k s^k$, where $p_k = \Pr\{Y_n = k\}$. Assume that the initial population size was $X_0 = 1$. Then for the first generation, the random variable X_1 will be just equal to Y : $X_1 = Y$

Note that for a branching process extinction probability condition (or condition of stationarity) is $g(s) = s$, where

8 Continuous-Time Markov Chain

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9 Continuous-Time Markov model

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10 Continuous-Time Markov Chain times

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11 Diffusion

Diffusion

12 More Diffusion

mode more more

13 Anomalous Diffusion

Anomalous

14 More Anomalous Diffusion

MORTE ANOMALOUS DIFFUSION

15 Diffusion in disordered systems

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